Inverse Probability Weighting and Marginal Structural Models

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Overview

This module introduces **Inverse Probability (IP) weighting** and **Marginal Structural Models (MSMs)** as fundamental approaches for estimating causal effects from observational data. Unlike stratification and outcome regression which we've studied previously, IP weighting creates a "pseudo-population" where treatment assignment becomes independent of measured confounders, allowing us to estimate causal effects by fitting simple associational models.

We'll explore this methodology using the motivating example of estimating the causal effect of smoking cessation on body weight gain using data from the National Health and Nutrition Examination Survey Data I Epidemiologic Follow-up Study (NHEFS). This real-world application demonstrates how IP weighting handles high-dimensional confounding and extends to complex scenarios involving continuous treatments, effect modification, and missing data.

The connection to marginal structural models provides a formal framework for understanding counterfactual outcomes and enables us to move beyond simple dichotomous treatments to more complex dose-response relationships.

Learning Objectives

By the end of this module, students will be able to:

1. **Explain** the conceptual foundation of IP weighting and how it creates a pseudo-population for causal inference (Knowledge/Comprehension)
2. **Distinguish** between nonstabilized and stabilized IP weights, identifying when each is appropriate and computing both by hand for simple examples (Analysis)
3. **Construct** and interpret marginal structural models for both dichotomous and continuous treatments, connecting model parameters to causal estimands (Application/Analysis)
4. **Implement** IP weighting estimation in R using `dagitty`, `ggdag`, and appropriate modeling functions, including diagnostic checks for positivity violations (Application)
5. **Evaluate** the assumptions required for valid IP weighting (exchangeability, positivity, consistency) and identify potential violations in real datasets (Evaluation)
6. **Design** and execute a complete IP weighting analysis including confounder selection, model specification, weight estimation, and sensitivity analysis (Synthesis)

R Packages

```
# Load required packages
pacman::p_load(
  tidyverse, # Data manipulation and visualization
  dagitty, # Creating and analyzing DAGs
  ggdag, # Visualizing DAGs with ggplot2
  MatchIt, # Propensity score methods
  WeightIt, # Flexible weighting framework
  cobalt, # Covariate balance assessment
  survey, # Survey-weighted analysis
  broom, # Tidy model outputs
  knitr, # Document formatting
```

```r
  kableExtra, # Enhanced tables
  latex2exp, # LaTeX expressions in plots
  MASS, # Statistical functions
  boot, # Bootstrap methods
  here, # File path management
  janitor, # Data cleaning
  patchwork # Combining plots
)

# Set theme for plots
theme_set(theme_minimal(base_size = 12))
```

Comprehensive Topic Explanation

The Fundamental Challenge

Observational data presents a persistent challenge: treatments are not randomly assigned. Instead, treatment assignment depends on observed and unobserved characteristics that may also influence the outcome. This creates **confounding** - a mixing of the causal effect we want to estimate with selection effects that bias our estimates.

Consider our motivating example: we want to estimate the causal effect of smoking cessation on weight gain using NHEFS data. Simply comparing mean weight gain between quitters and non-quitters gives us:

- Quitters: $\hat{\mathbb{E}}[Y|A=1] = 4.5$ kg
- Non-quitters: $\hat{\mathbb{E}}[Y|A=0] = 2.0$ kg
- Naive difference: $4.5 - 2.0 = 2.5$ kg

But this associational difference $\mathbb{E}[Y|A=1] - \mathbb{E}[Y|A=0]$ is not the causal difference $\mathbb{E}\left[Y^{a=1}\right] - \mathbb{E}\left[Y^{a=0}\right]$ we seek, because quitters and non-quitters differ systematically in characteristics that affect weight gain (Hernán & Robins, 2020).

IP Weighting: The Core Intuition

IP weighting addresses confounding by creating a **pseudo-population** where treatment assignment is independent of measured confounders. The key insight is to weight each individual by the inverse probability of receiving the treatment they actually received.

For individual i with treatment A_i and confounders L_i, the IP weight is:

$$W_i^A = \frac{1}{f(A_i|L_i)} \tag{1}$$

where $f(A_i|L_i)$ is the conditional probability (or density) of receiving treatment A_i given confounders L_i.

Why Does This Work?

In the weighted pseudo-population, individuals who received an unlikely treatment (given their confounders) are up-weighted, while those who received a likely treatment are down-weighted. This rebalancing creates a population where:

1. Treatment assignment is independent of confounders: $A \perp\!\!\!\perp L$
2. The standardized mean equals the counterfactual mean: $\mathbb{E}_{ps}[Y|A = a] = \mathbb{E}[Y^a]$
3. Association equals causation: We can estimate causal effects using simple associational models

Mathematical Foundation

The pseudo-population has two crucial properties (Hernán & Robins, 2020):

Property 1 (Independence): In the pseudo-population, $A \perp\!\!\!\perp L$.

Property 2 (Preservation of Counterfactual Means):

$$\mathbb{E}_{ps}[Y|A=a] = \sum_{l} \mathbb{E}[Y|A=a, L=l]\, \mathbb{P}(L=l) = \mathbb{E}[Y^a] \qquad (2)$$

The second equality holds under conditional exchangeability $Y^a \perp\!\!\!\perp A|L$.

From Nonstabilized to Stabilized Weights

The basic IP weights $W^A = 1/f(A|L)$ are called **nonstabilized weights**. While they create the desired pseudo-population, they can have high variability, leading to inefficient estimates.

Stabilized weights address this by including the marginal treatment probability in the numerator:

$$SW^A = \frac{f(A)}{f(A|L)} \qquad (3)$$

where $f(A)$ is the marginal probability of treatment.

For dichotomous treatment:

- Treated individuals: $SW^A = \frac{\mathbb{P}(A=1)}{\mathbb{P}(A=1|L)}$
- Untreated individuals: $SW^A = \frac{\mathbb{P}(A=0)}{\mathbb{P}(A=0|L)}$

Stabilized weights preserve the causal interpretation while typically yielding more efficient estimates, especially in complex models.

Extension to Marginal Structural Models

A **marginal structural model** specifies the relationship between treatment and counterfactual outcomes:

$$\mathbb{E}[Y^a] = g(a; \beta) \qquad (4)$$

where $g(\cdot)$ is a known function (e.g., linear, logistic) and β are parameters to be estimated.

For example, a linear MSM for dichotomous treatment:

$$\mathbb{E}[Y^a] = \beta_0 + \beta_1 a$$

Here, $\beta_1 = \mathbb{E}\left[Y^{a=1}\right] - \mathbb{E}\left[Y^{a=0}\right]$ is the average causal effect.

The key insight is that parameters of the MSM can be estimated by fitting the corresponding associational model to the IP-weighted pseudo-population:

$$\mathbb{E}[Y|A] = \theta_0 + \theta_1 A$$

Under our assumptions, $\hat{\theta}_1$ provides a consistent estimate of β_1.

Handling Complex Treatments

Continuous Treatments

For continuous treatments, IP weights require estimating probability density functions rather than probabilities. This typically involves assuming a distributional form (e.g., normal) and estimating parameters:

$$f(A|L) \sim \mathcal{N}(\mu_L, \sigma^2)$$

where $\mu_L = \mathbb{E}[A|L]$ and σ^2 are estimated via regression.

Polytomous Treatments

For treatments with multiple categories, we use multinomial models to estimate $\mathbb{P}(A = a|L)$ for each treatment level a.

Assumptions for Valid Inference

IP weighting requires the same fundamental assumptions as other causal inference methods:

1. **Exchangeability**: $Y^a \perp\!\!\!\perp A|L$ (no unmeasured confounding)
2. **Positivity**: $0 < \mathbb{P}(A = a|L = l) < 1$ for all relevant (a, l)

3. **Consistency**: $Y_i = Y_i^{a_i}$ when $A_i = a_i$

Additionally, we need:

4. **Correct model specification** for the propensity score $f(A|L)$

Violations of positivity are particularly problematic for IP weighting, as they lead to extreme weights and unstable estimates.

Advantages and Limitations

Advantages:

- Separates design (weight estimation) from analysis phases
- Natural extension to time-varying treatments
- Transparent handling of effect modification
- Robust to outcome model misspecification

Limitations:

- Sensitive to positivity violations
- Requires correct propensity score model
- Can be inefficient with extreme weights
- Challenging for continuous treatments

Intuitive Clarifications of Challenging Ideas

The Pseudo-Population Metaphor

Think of IP weighting as creating a **hypothetical randomized experiment**. In this imaginary study:

- Everyone has the same probability of receiving each treatment level
- We've "removed the arrows" from confounders to treatment in our causal diagram
- The treatment assignment mechanism is now random rather than systematic

Visualization Aid

Imagine we have 1000 people in our original study. After IP weighting, we might have an effective sample size that looks like 2000 people, but these aren't "new" people—they're weighted copies that rebalance the original sample to look like what we'd get from random assignment.

Why "Inverse" Probability?

The term "inverse probability" often confuses students. Here's the intuition:

- If someone has a **high** probability of receiving their observed treatment, they represent many similar people → **low** weight (down-weight the common case)
- If someone has a **low** probability of receiving their observed treatment, they represent few similar people → **high** weight (up-weight the rare case)

Example: Consider a smoker who quits despite having characteristics that typically predict continued smoking. This person gets a high weight because they represent the many similar smokers who didn't quit but would have gained weight if they had.

The Stabilization Concept

Think of stabilized weights as **variance control**:

- Nonstabilized weights: $1/\,\mathbb{P}(\text{treatment}|\text{confounders})$
- Stabilized weights: $\mathbb{P}(\text{treatment})\,/\,\mathbb{P}(\text{treatment}|\text{confounders})$

The numerator acts as a **stabilizing factor** that prevents weights from becoming too extreme while preserving the causal interpretation.

Key Insight: Stabilization Preserves the Population

Stabilized weights maintain the original population size in expectation: $\mathbb{E}[SW^A] = 1$. This means our effective sample size stays approximately the same, unlike nonstabilized weights where $\mathbb{E}[W^A] > 1$.

Marginal Vs Conditional Models

A common source of confusion is the distinction between marginal structural models and conditional models:

- **Conditional model**: $\mathbb{E}[Y|A,L] = \alpha_0 + \alpha_1 A + \alpha_2^T L$
  - Includes confounders as covariates
  - α_1 is the effect adjusting for L
- **Marginal structural model**: $\mathbb{E}[Y^a] = \beta_0 + \beta_1 a$
  - No confounders in the model
  - β_1 is the marginal (population-average) causal effect

The MSM gives us the effect we'd see if we intervened on the entire population, while the conditional model gives us the effect within strata of L. This is a key distinction.

Common Misconceptions

Misconception 1: "IP Weighting Eliminates confounding"

Reality: IP weighting creates a pseudo-population where confounding is absent, but this depends on measuring all relevant confounders and correct model specification.

Misconception 2: "Heavier Weights Are Always better"

Reality: Extreme weights often indicate positivity violations or model misspecification. Weights should be examined for reasonableness.

⚠ Misconception 3: "Stabilized Weights Are Always preferred"

Reality: For saturated models (like simple dichotomous treatment), stabilized and nonstabilized weights yield identical point estimates. Stabilized weights primarily help with efficiency in complex models.

Simple, Hand-Calculation Numerical Example(s)

Let's work through IP weighting with a minimal example to build intuition.

Worked Example: Smoking and Weight Gain

Consider 6 individuals with data on:

- A: Quit smoking (1=yes, 0=no)
- Y: Weight gain (kg)
- L: Age group (1=older, 0=younger)

Person	L (Age)	A (Quit)	Y (Weight)
1	0	0	1
2	0	1	4
3	1	0	2
4	1	0	1
5	1	1	3
6	1	1	5

Step 1: Calculate Crude Association

$$\hat{\mathbb{E}}\left[Y|A=1\right] = \frac{4+3+5}{3} = 4.0 \text{ kg}$$

$$\hat{\mathbb{E}}\left[Y|A=0\right] = \frac{1+2+1}{3} = 1.33 \text{ kg}$$

$$\text{Crude difference} = 4.0 - 1.33 = 2.67 \text{ kg}$$

Step 2: Estimate Propensity Scores

Calculate $\hat{\mathbb{P}}(A = 1|L = l)$ for each age group:

- Younger $(L = 0)$: 1 quit out of 2 total → $\hat{\mathbb{P}}(A = 1|L = 0) = 0.5$
- Older $(L = 1)$: 2 quit out of 4 total → $\hat{\mathbb{P}}(A = 1|L = 1) = 0.5$

Step 3: Calculate Nonstabilized IP Weights

For each person: $W_i^A = 1/\hat{\mathbb{P}}(A = A_i|L = L_i)$

Person	L	A	$\hat{\mathbb{P}}(A = A_i \| L = L_i)$	W_i^A
1	0	0	0.5	2.0
2	0	1	0.5	2.0
3	1	0	0.5	2.0
4	1	0	0.5	2.0
5	1	1	0.5	2.0
6	1	1	0.5	2.0

Step 4: Estimate Causal Effect

Using weighted means in the pseudo-population:

$$\hat{\mathbb{P}}_{ps}(Y|A = 1) = \frac{\sum_{A_i=1} W_i^A Y_i}{\sum_{A_i=1} W_i^A} = \frac{2.0(4) + 2.0(3) + 2.0(5)}{2.0 + 2.0 + 2.0} = \frac{24}{6} = 4.0$$

$$\hat{\mathbb{P}}_{ps}(Y|A = 0) = \frac{\sum_{A_i=0} W_i^A Y_i}{\sum_{A_i=0} W_i^A} = \frac{2.0(1) + 2.0(2) + 2.0(1)}{2.0 + 2.0 + 2.0} = \frac{8}{6} = 1.33$$

$$\text{IP weighted difference} = 4.0 - 1.33 = 2.67 \text{ kg}$$

Step 5: Compare with Standardization

For comparison, let's calculate the standardized estimate:

$$\hat{\mathbb{E}}\left[Y^{a=1}\right] = \sum_l \hat{\mathbb{E}}\left[Y|A=1, L=l\right] \hat{\mathbb{P}}(L=l)$$

$$= \hat{\mathbb{E}}\left[Y|A=1, L=0\right] \times 0.33 + \hat{\mathbb{E}}\left[Y|A=1, L=1\right] \times 0.67$$

$$= 4.0 \times 0.33 + 4.0 \times 0.67 = 4.0$$

Similarly, $\hat{\mathbb{E}}\left[Y^{a=0}\right] = 1.33$.

The standardized and IP weighted estimates agree, as expected!

Step 6: Stabilized Weights

Let's calculate stabilized weights. First, estimate marginal probabilities:

$$\hat{\mathbb{P}}(A=1) = 3/6 = 0.5$$

$$\hat{\mathbb{P}}(A=0) = 3/6 = 0.5$$

Stabilized weights: $SW_i^A = \hat{\mathbb{P}}([)A = A_i]/\hat{\mathbb{P}}([)A = A_i|L = L_i]$

Person	$\hat{\mathbb{P}}([)A = A_i]$	SW_i^A
1	0.5	1.0
2	0.5	1.0
3	0.5	1.0
4	0.5	1.0
5	0.5	1.0
6	0.5	1.0

Since propensity scores equal marginal probabilities, stabilized weights all equal 1, and we get the same causal effect estimate.

Hand Calculation with Confounding

Example with True Confounding

Modify the data to show confounding:

Person	L	A	Y	Notes
1	0	0	3	Young non-quitter
2	0	1	5	Young quitter
3	1	0	1	Old non-quitter
4	1	0	0	Old non-quitter
5	1	1	2	Old quitter
6	1	1	3	Old quitter

Now younger people (who gain more weight naturally) are equally likely to quit, but older people (who gain less weight naturally) are also equally likely to quit.

Crude Association

- Quitters: $(5 + 2 + 3)/3 = 3.33$ kg
- Non-quitters: $(3 + 1 + 0)/3 = 1.33$ kg
- Crude difference: 2.0 kg

Propensity Scores

Same as before: $\hat{\mathbb{P}}([)\, A = 1|L = 0] = \hat{\mathbb{P}}([)\, A = 1|L = 1] = 0.5$

IP Weighted Analysis

All weights are 2.0, giving the same 2.0 kg effect.

Standardized Analysis

$$\hat{\mathbb{E}}\left[Y^{a=1}\right] = EY|A=1, L=0 \times 0.33 + EY|A=1, L=1 \times 0.67$$

$$= 5.0 \times 0.33 + 2.5 \times 0.67 = 3.33$$

$$\hat{\mathbb{E}}\left[Y^{a=0}\right] = 3.0 \times 0.33 + 0.5 \times 0.67 = 1.33$$

Causal effect: $3.33 - 1.33 = 2.0$ kg, matching IP weighting.

R Code Examples & Interpretation

Let's implement IP weighting using simulated data that mimics the NHEFS structure.

Simulating NHEFS-like Data

```
set.seed(12345)
n <- 1500

# Generate baseline confounders
age <- rnorm(n, mean = 43, sd = 12)
age <- pmax(25, pmin(age, 74)) # Constrain to 25-74
sex <- rbinom(n, 1, 0.5) # 1 = female
race <- rbinom(n, 1, 0.15) # 1 = non-white
education <- sample(1:5, n, replace = TRUE, prob =
c(0.1, 0.2, 0.3, 0.25, 0.15))
weight_baseline <- rnorm(n, mean = 70 + 5 * sex, sd = 15)
cigarettes_day <- rpois(n, lambda = 20)
years_smoking <- pmax(1, rnorm(n, mean = 25, sd = 8))
exercise <- sample(1:3, n, replace = TRUE, prob = c(0.4,
0.4, 0.2))
activity <- sample(1:3, n, replace = TRUE, prob = c(0.1,
0.8, 0.1))
```

```r
# Create treatment model (quit smoking)
# Older, more educated, lower baseline smoking more
likely to quit
quit_logit <- -2 +
  0.03 * age +
  0.5 * education -
  0.02 * cigarettes_day +
  0.4 * sex -
  0.01 * weight_baseline +
  0.01 * years_smoking
quit_prob <- plogis(quit_logit)
quit <- rbinom(n, 1, quit_prob)

# Create outcome model (weight gain)
# Quitting causes weight gain, but also depends on
confounders
weight_gain <- 1 +
  2.5 * quit +
  0.02 * age -
  0.8 * sex +
  0.3 * race +
  0.1 * education +
  0.01 * weight_baseline -
  0.05 * cigarettes_day +
  rnorm(n, mean = 0, sd = 2)

# Combine into dataset
nhefs_sim <- tibble(
  id = 1:n,
  quit = quit,
  weight_gain = weight_gain,
  age = age,
  sex = sex,
  race = race,
  education = education,
  weight_baseline = weight_baseline,
```

```
  cigarettes_day = cigarettes_day,
  years_smoking = years_smoking,
  exercise = exercise,
  activity = activity
)

# Check crude association
crude_effect <- nhefs_sim |>
  group_by(quit) |>
  summarise(mean_wt = mean(weight_gain), .groups =
  'drop') |>
  pull(mean_wt) |>
  diff()

cat("Crude association:", round(crude_effect, 2),
"kg\n")
```

```
Crude association: 2.61 kg
```

```
cat("True causal effect: 2.5 kg\n")
```

```
True causal effect: 2.5 kg
```

Creating and Visualizing the DAG

```
# Create DAG
smoking_dag <- dagitty(
  'dag {
  Age [pos="0,0"]
  Sex [pos="1,0"]
  Education [pos="2,0"]
  Baseline_Weight [pos="0,1"]
  Cigarettes [pos="1,1"]
  Years_Smoking [pos="2,1"]
```

```
  Quit_Smoking [pos="1,2"]
  Weight_Gain [pos="2,2"]

  Age -> Quit_Smoking
  Sex -> Quit_Smoking
  Education -> Quit_Smoking
  Baseline_Weight -> Quit_Smoking
  Cigarettes -> Quit_Smoking
  Years_Smoking -> Quit_Smoking

  Age -> Weight_Gain
  Sex -> Weight_Gain
  Education -> Weight_Gain
  Baseline_Weight -> Weight_Gain
  Cigarettes -> Weight_Gain

  Quit_Smoking -> Weight_Gain
}'
)

# Visualize DAG
ggdag(smoking_dag, layout = "circle") +
  theme_dag_blank() +
  labs(
    title = "Causal DAG: Smoking Cessation -> Weight
    Gain",
    subtitle = "Multiple confounders create backdoor
    paths"
  )
```

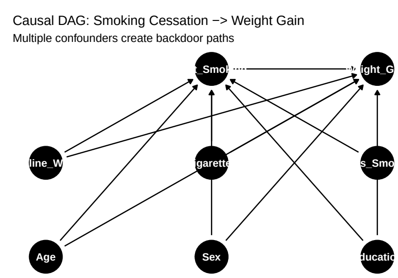

Figure 1: Causal DAG for smoking cessation and weight gain

```
# Check for adjustment sets
adjustmentSets(
  smoking_dag,
  exposure = "Quit_Smoking",
  outcome = "Weight_Gain",
```

```r
  type = "minimal"
)
```

```
{ Age, Baseline_Weight, Cigarettes, Education, Sex }
```

```r
# Check implied independence statements
impliedConditionalIndependencies(smoking_dag)
```

```
Age _||_ Bs_W
Age _||_ Cgrt
Age _||_ Edct
Age _||_ Sex
Age _||_ Yr_S
Bs_W _||_ Cgrt
Bs_W _||_ Edct
Bs_W _||_ Sex
Bs_W _||_ Yr_S
Cgrt _||_ Edct
Cgrt _||_ Sex
Cgrt _||_ Yr_S
Edct _||_ Sex
Edct _||_ Yr_S
Sex _||_ Yr_S
Wg_G _||_ Yr_S | Age, Bs_W, Cgrt, Edct, Qt_S, Sex
```

Propensity Score Estimation

```r
# Fit logistic model for Pr(A = 1 | L)
ps_model <- glm(
  quit ~
    age +
      I(age^2) +
      sex +
      race +
```